Optimization Methods for Machine Learning and Engineering

Lecture 1 – Introduction, Convexity & Gradient Descent

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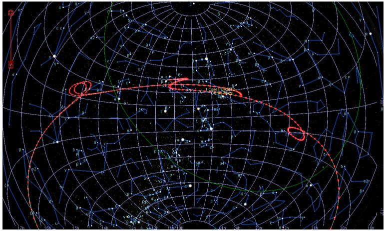
Agenda

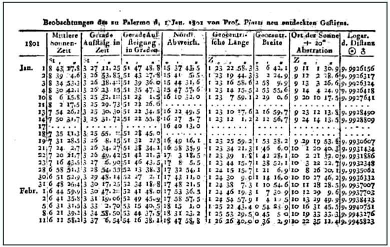
1. Motivation and Historical Perspective
2. Course Organization
3. Math Notation and Analysis Refresher
4. Convexity
5. Gradient Descent

Motivation and Historical Perspective

- In the early 19th century, Giuseppe Piazzi published ***observations*** of the newly discovered ***dwarf planet Ceres***
- C.F. Gauß was able to ***compute the orbit of Ceres from the observations*** [Gauss1823]. It was found at the predicted location many weeks after sight had been lost.
- To fit the model parameters from (noisy) observations, Gauß invented the ***Least Squares Method***
- Fitting of model parameters to reduce the prediction error is also known as ***Empirical Risk Minimization***

Many ***Machine Learning Techniques***
solve an ***Optimization Problem***





Beobachtungen des zu Palermo d. 1 Jan. 1801 von Prof. *Piazzi* neu entdeckten Gestirns.

1801	Mittlere Sonnen-Zeit	Gerade Aufstäig in Zeit	Gerade Aufsteigung in Graden	Nördl. Abweich.	Geocentrische Länge	Geocentr. Breite	Ort der Sonne + 20" Aberration	Logar. d. Distanz ⊙ ♁
	St ′ ″	St ′ ″	° ′ ″	° ′ ″	Z ° ′ ″	° ′ ″	Z ° ′ ″	
Jan. 1	8 43 27,8	3 27 11,25	51 47 48,8	15 37 43,5	1 23 22 58,3	3 6 42,1	9 11 1 30,9	9,9926156
2	8 39 4,6	3 26 53,85	51 43 27,8	15 41 5,5	1 23 19 44,3	3 2 24,9	9 12 2 28,6	9,9926317
3	8 34 53,3	3 26 38,4[illegible]	51 39 36,0	15 44 31,6	1 23 16 58,6	2 58 9,9	9 13 3 26,6	9,9926324
4	8 30 42,1	3 26 23,16	51 35 47,3	15 47 57,6	1 23 14 15,5	2 53 55,6	9 14 4 24,9	9,9926418
10	8 6 15,8	3 25 32,1[illegible]	51 28 1,5	16 10 32,0	1 23 7 59,1	2 29 0,6	9 20 10 17,5	9,9927641
11	8 2 17,5	3 25 29,73	51 22 26,0	. . . . . .	. . . . . .	. . . . .	. . . . . .	. . . . .
13	7 54 26,2	3 25 30,30	51 22 34,5	16 22 49,5	1 23 10 27,6	2 16 59,7	9 23 12 13,8	9,9928490
14	7 50 31,7	3 25 31,72	51 22 55,8	16 27 5,7	1 23 12 1,2	2 12 56,7	9 24 14 13,5	9,9928809
17	. . . . . .	. . . . . .	. . . . . .	16 40 13,0	. . . . . .	. . . . .	. . . . . .	. . . . .
18	7 35 11,3	3 25 55,[illegible]	51 28 45,0	. . . . . .	. . . . . .	. . . . .	. . . . . .	. . . . .
19	7 31 28,5	3 26 8,15	51 32 2,3	16 49 16,1	1 23 25 59,2	1 53 38,2	9 29 19 53,8	9,9930607
21	7 24 2,7	3 26 34,27	51 38 34,1	16 58 35,9	1 23 34 21,3	1 46 6,0	10 1 20 40,3	9,9931434
22	7 20 21,7	3 26 49,42	51 42 21,3	17 3 18,5	1 23 39 1,8	1 42 28,1	10 2 21 32,0	9,9931886
23	7 16 43,5	3 27 6,90	51 46 43,5	17 8 5,5	1 23 44 15,7	1 38 52,1	10 3 22 22,7	9,9932348
28	6 58 51,3	3 28 54,53	52 13 38,3	17 32 54,1	1 24 15 15,7	1 21 6,9	10 8 26 20,1	9,9935062
30	6 51 52,9	3 29 48,14	52 27 2,1	17 43 11,0	1 24 30 9,0	1 14 16,0	10 10 27 46,2	9,9936332
31	6 48 26,4	3 30 17,25	52 34 18,8	17 48 21,5	1 24 38 7,3	1 10 54,6	10 11 28 28,5	9,9937007
Febr. 1	6 44 59,9	3 30 47,2[illegible]	52 41 48,0	17 53 36,3	1 24 46 19,3	1 7 30,9	10 12 29 9,6	9,9937703
2	6 41 35,8	3 31 19,06	52 49 45,9	17 58 57,5	1 24 54 57,9	1 4 1,5	10 13 29 49,9	9,9938423
5	6 31 31,5	3 33 2,70	53 15 40,5	18 15 1,0	1 25 22 43,4	0 54 23,9	10 16 31 45,5	9,9940751
8	6 21 39,2	3 34 58,50	53 44 37,5	18 31 23,2	1 25 53 29,5	0 45 5,0	10 19 33 33,3	9,9943276
11	6 11 58,2	3 37 6,54	54 16 38,1	18 47 58,8	1 26 26 40,0	0 36 2,9	10 22 35 11,4	9,9945823

Images taken from [Mansfield16]

Stigler's Diet

- In the 1930s, the US Army was interested in ***low-cost diets*** that keep soldiers alive/healthy
  - Daily nutrition requirements
  - Nutrition content of 77 food types and their price
  - What is the cheapest combination of food that meets the requirements?
- ***Previous attempts*** gave diets for about ***$100 a year***
- George Stigler used math. optimization [Stigler45]
  - Nine clercs worked for ***120 person days*** on the calculations
  - The solution had an ***annual cost of $39.93***
- Stigler's diet belongs to a class of optimization problems called ***Linear Programs (LP)***
- Today, LP are solved routinely with standard algorithms. Stigler's diet is solved in milliseconds.

TABLE 1. DAILY ALLOWANCES OF NUTRIENTS FOR A MODERATELY ACTIVE MAN (weighing 154 pounds)*

Nutrient	Allowance
Calories	3,000 calories
Protein	70 grams
Calcium	.8 grams
Iron	12 milligrams
Vitamin A	5,000 International Units
Thiamine (B_1)	1.8 milligrams
Riboflavin (B_2 or G)	2.7 milligrams
Niacin (Nicotinic Acid)	18 milligrams
Ascorbic Acid (C)	75 milligrams

* National Research Council, *Recommended Dietary Allowances*, Reprint and Circular Series No. 115, January, 1943.

Stigler's Solution

Food	Annual Quantities	Annual Cost
Wheat Flour	167.8 kg	$13.33
Evaporated Milk	57 cans	$3.84
Cabbage	50.3 kg	$4.11
Spinach	10.4 kg	$1.85
Dried Navy Beans	129.3 kg	$16.80
Total Annual Cost		**$39.93**

George Dantzig wrote a humorous account on applying optimized diets in real life: George B Dantzig. "The diet problem". In: *Interfaces* 20.4 (1990), pp. 43–47

Cost Optimization under Requirement Constraints

$$\begin{aligned} \text{minimize} \quad & \mathrm{cost}(\boldsymbol{x}) \\ \text{subject to} \quad & \mathrm{performance}(\boldsymbol{x}) \geq \mathrm{requirements} \end{aligned}$$

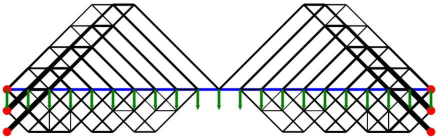
Performance Optimization under Cost Constraints

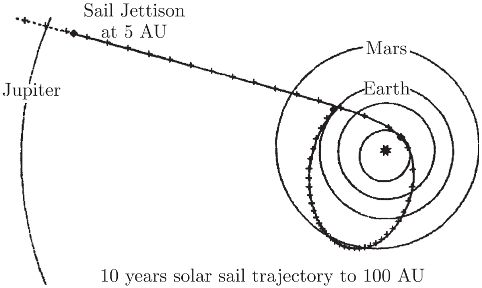
$$\begin{aligned} \text{maximize} \quad & \mathrm{performance}(\boldsymbol{x}) \\ \text{subject to} \quad & \mathrm{cost}(\boldsymbol{x}) \leq \mathrm{budget} \end{aligned}$$

Cost-Performance Tradeoff

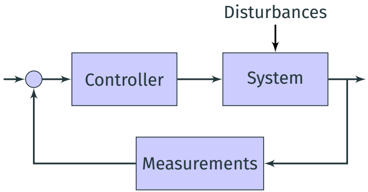
$$\text{maximize} \quad \mathrm{performance}(\boldsymbol{x}) - \alpha\, \mathrm{cost}(\boldsymbol{x})$$

For an appropriate solution space $\boldsymbol{x} \in X$.





[Sauer1999]

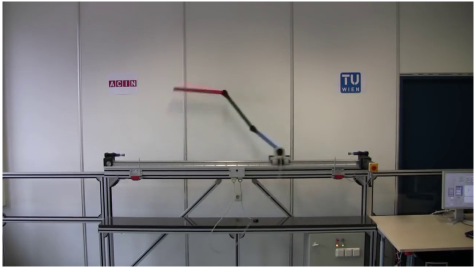


- ***Closed-Loop Control***, also called ***Feedback Control***
- Originally developed as ***Cybernetics***, today ***Control Theory*** and ***Systems Theory***
- ***Model Predictive Control:*** Control values are the result of an optimization problem. To be solved many times per second.

Application Example: Autonomous RC Car Control

Source: `https://control.ee.ethz.ch/research/team-projects/autonomous-rc-car-racing.html`

Application Example: Inverted Pendulum



Source: `https://www.acin.tuwien.ac.at/en/`

Optimization for Computer Vision

Stereo Matching

Image Denoising

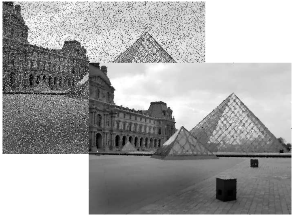


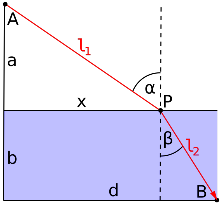
Image Deblurring

- An ***Energy Function*** describes the solution quality (similar to loss functions in machine learning)
- Optimization Problems in Computer Vision are ***very high-dimensional!***
- Non-convex energy functions are approximated or solved iteratively

See e.g. [Komodakis07; Pock09; Chambolle16] for recent results in this domain.

Optimization in Nature

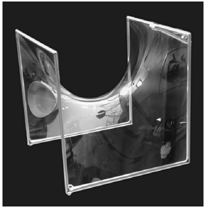
Light Travels the Shortest Path (Fermat's Principle)



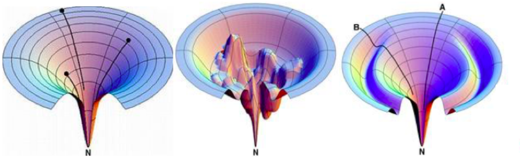
Optimization as a ***Principle of Nature***

- Minimum Energy States
- Principle of Least Action
    - Light travels the shortest path
    - Classical Mechanics
- Goal-Directed Active Behavior

Soap Film of Minimum Surface



Energy Minimization for Protein Folding

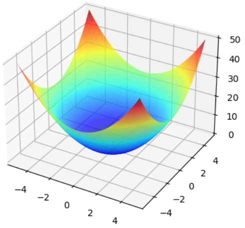


[Dill1997]

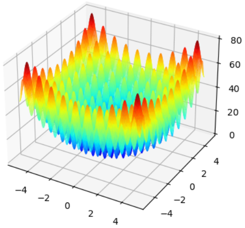
$$\boldsymbol{x}^* = \arg\min_{\boldsymbol{x} \in X} \; f(\boldsymbol{x})$$

- Minimize the ***objective function*** f ...
    - on the ***domain*** X ...
    - to find the ***minimizer*** x^*.
- If f is ***strictly convex*** on X, then the ***minimizer is unique***

Strictly convex function have only ***one minimizer.***
To find it, ***just walk downhill!***



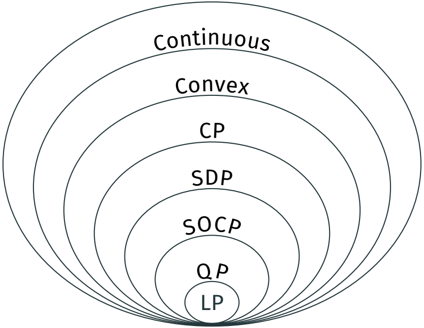
Convex: Easy



Non-Convex: Difficult

What can be expressed in terms of convex functions is surprisingly rich.
Large “bag of tricks” to approximate a problem with a convex function.

The Hierarchy of Optimization Problems



Abbreviations

CP: Conic Programming

SDP: Semi-Definite Programming

SOCP: Second-Order Cone Programming

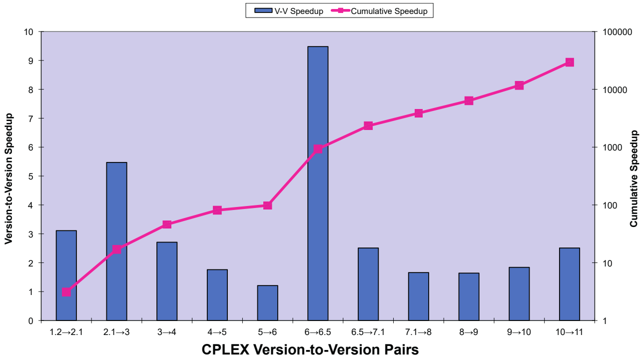
QP: Quadratic Programming

LP: Linear Programming

We will get to know these gradually throughout the course.

- More general problem classes include ***subsumed problem classes***. Eg. $LP \subset QP$
- ***Solution techniques*** for a more general problem class also ***solve the subsumed problem classes***
- But: ***Specialized techniques*** are usually faster as they can ***exploit stronger assumptions***

The Success of Optimization Software Packages



- ***Speed-up by many orders of magnitude*** in the 1990s period [Bixby2012]
  - Factor 30,000 by better algorithms (for a benchmark set of the most challenging problems)
  - Factor 1,000 by better hardware
- Many ***domain-specific*** solution techniques were ***replaced by generic*** optimization tools

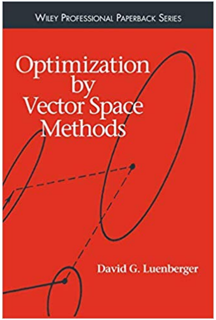
Course Organization

Syllabus

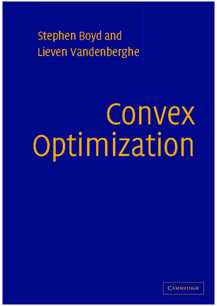
1. **Introduction, Convexity and Gradient Descent**
2. **Newton's Method**
3. **Inequality Constrained Optimization**
4. **Equality Constrained Optimization**
5. **Applications: Mechanical Design, Model-Predictive Control, Optimization in Finance**
6. **Automatic Differentiation and Neural Networks**
7. **Vector Spaces, Norms and the Projection Theorem**
8. **Fast First-Order Optimization**
9. **Duality and Primal-Dual Algorithms**
10. **SVM and the Reproducing Kernel Hilbert Space**
11. **Conic Programming**
12. **Alternating Methods and the EM Algorithm**
13. **Applications: Graph Problems, Computer Vision and Generalized Low-Rank Models**
14. **Gradient-Free and Non-Convex Optimization**

- The lectures are accompanied by a ***weekly exercise sessions***
- ***Exercise sheets and Jupyter notebooks*** are released together with the lecture recordings and slides
- It is useful (but not required) to ***bring a laptop to the exercise sessions***

Main References



David G Luenberger. *Optimization by Vector Space Methods*. John Wiley & Sons, 1969



Stephen Boyd and Lieven Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004

The relevant material is covered in the lectures. You don't have to buy these books to follow the class.

Optimization at KIT

- **Kontinuierliche Optimierung**
  Institute for Operations Research, Prof. Stein
  `https://kop.ior.kit.edu/index.php`
- **Naturanaloge und verteilte Optimierungsverfahren**
  Institute for Applied Informatics and Formal Description Methods, Pradyumn Shukla
  `https://aifb.kit.edu/index.php/Lehre/Vorlesung_Naturanaloge_und_verteilte_Optimierungsverfahren/en`
- **Optimierungstheorie**
  Institute for Applied and Numerical Mathematics, Prof. Griesmaier
  `https://www.math.kit.edu/ianmip/lehre/opttheo2020s/de`
- **Optimization of Dynamic Systems**
  Institute of Control Systems, Prof. Hohmann
  `https://www.irs.kit.edu/Lectures_ODS.php`
- ... a few more courses ... and a lot more applications in the different disciplines!

> This course is different. Focus on the intersection of Optimization Theory,
> ***Machine Learning*** and ***Engineering Applications.***

Math Notation and Analysis Refresher

Linear Algebra

- a: Scalar
- $\boldsymbol{a}$: Vector
- $\mathbf{A}$: Matrix
- $\mathbf{A}^\top$: Matrix Transpose
- $\mathbf{A}^{-1}$: Matrix Inverse

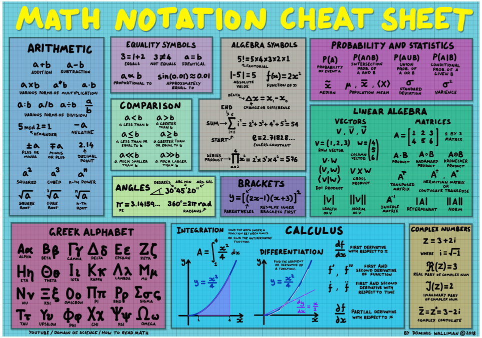
Set Theory

- $a \in A$: a is an element of set A
- $A \subseteq B$: A is a subset of (or equal to) B
- $A \cup B$: The union of A and B
- $A \cap B$: The intersection of A and B
- $\varnothing = \{\,\}$: The empty set
- $\{x \mid x^2 = 1\}$: The set of all x with $x^2 = 1$

Abbreviations

- iff: if and only if
- rhs/lhs: right-hand-side, left-hand-side

Basic Math Cheat Sheet



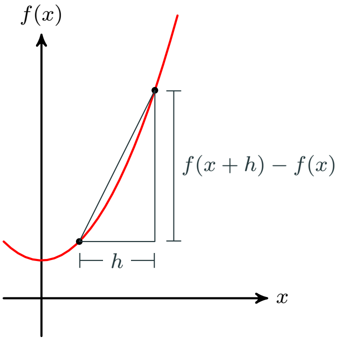
https://www.youtube.com/watch?v=Kp2bYWRQylk

Come back to this slide when the need arises.
Or just ask during lectures / tutorials.

Definition

Let X be an open subset of $\mathbb{R}$ and let $f : X \to \mathbb{R}$. Then f is differentiable at $x \in X$ with derivative $\frac{d}{dx} f(x)$ if the following limit exists:

$$\frac{d}{dx} f(x) = \lim_{h \to 0} \frac{1}{h} \big(f(x+h) - f(x) \big)$$



Example for $f(x) = 2x + 3x^2$

$$\begin{aligned}
\tfrac{d}{dx} f(x) &= \lim_{h \to 0} \tfrac{1}{h} \big(2(x+h) + 3(x+h)^2 - 2x - 3x^2 \big) \\
&= \lim_{h \to 0} \tfrac{1}{h} \big(2(x+h) + 3(x^2 + 2xh + h^2) - 2x - 3x^2 \big) \\
&= \lim_{h \to 0} \tfrac{1}{h} \big(2h + 3(2xh + h^2) \big) \\
&= \lim_{h \to 0} \big(2 + 6x + 3h \big) = 2 + 6x
\end{aligned}$$

Gradient and Partial Derivatives

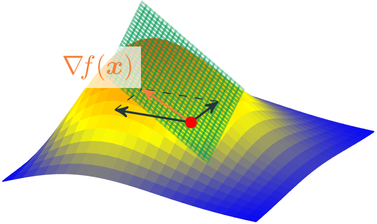
A function of several variables can be written as $f(x_1, x_2)$, etc. Often times, we abbreviate multiple arguments in a single vector as $f(\boldsymbol{x})$.

Let a function $f : \mathbb{R}^n \to \mathbb{R}$. The ***gradient of*** f is the ***column vector of partial derivatives***

$$\nabla f(\boldsymbol{x}) := \begin{pmatrix} \frac{\partial f(\boldsymbol{x})}{\partial x_1} \\ \vdots \\ \frac{\partial f(\boldsymbol{x})}{\partial x_n} \end{pmatrix}$$

Suppose now a function $g(\boldsymbol{x}, \boldsymbol{y})$ with signature $g : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$. Its ***derivative with respect to just*** x is written as $\nabla_{\boldsymbol{x}} g(\boldsymbol{x}, \boldsymbol{y})$.

Gradient and Tangent Plane / 1st Degree Taylor Expansion



$$\tau^1_{\boldsymbol{x}}(\boldsymbol{y}) = f(\boldsymbol{x}) + (\boldsymbol{y} - \boldsymbol{x})^\top \nabla f(\boldsymbol{x})$$

The Hessian Matrix

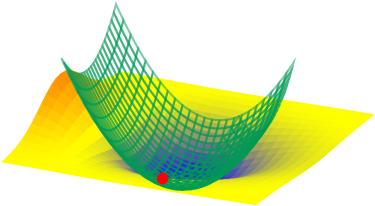
Let $f : \mathbb{R}^n \to \mathbb{R}$ twice differentiable. Its ***second (partial) derivatives*** make up the ***Hessian Matrix*** $\nabla^2 f(\boldsymbol{x})$:

$$\nabla^2 f(\boldsymbol{x}) := \begin{pmatrix} \frac{\partial^2 f(\boldsymbol{x})}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f(\boldsymbol{x})}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f(\boldsymbol{x})}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f(\boldsymbol{x})}{\partial x_n \partial x_n} \end{pmatrix}$$

- The ***order of differentiation does not matter*** if the function has continuous second (higher-order) partial derivatives (Schwarz's Theorem)
- Then the ***Hessian is symmetric***

$$\nabla^2 f(\boldsymbol{x}) = [\nabla^2 f(\boldsymbol{x})]^\top$$

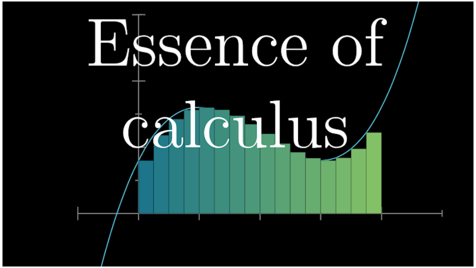
2nd Degree Taylor Expansion



$$\tau^2_{\boldsymbol{x}}(\boldsymbol{y}) = f(\boldsymbol{x}) + (\boldsymbol{y} - \boldsymbol{x})^\top \nabla f(\boldsymbol{x}) + \frac{1}{2}(\boldsymbol{y} - \boldsymbol{x})^\top \left[\nabla^2 f(\boldsymbol{x})\right](\boldsymbol{y} - \boldsymbol{x})$$

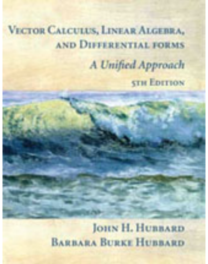
Good references if you want to refresh even more...

ThreeBlueOneBrown on Youtube



https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr

The Book by Hubbard and Hubbard



John H Hubbard and Barbara Burke Hubbard. *Vector Calculus, Linear Algebra, and Differential Forms: A Unified Approach*. Matrix Editions, 2015

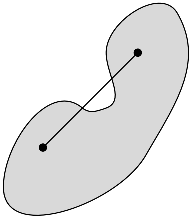
...or your old course material on Analysis for that matter!

Convexity

Convex Set
A set X is convex iff the line segment between any two points $\boldsymbol{x}, \boldsymbol{y} \in X$ remains in X.

$$\forall \boldsymbol{x}, \boldsymbol{y} \in X,\ 0 \leq \alpha \leq 1,\ \alpha \boldsymbol{x} + (1-\alpha)\boldsymbol{y} \in X$$

Non-Convex Set

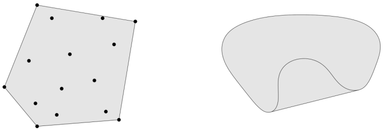


Convex Hull
The convex hull of a set $X = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_k\}$, denoted $\mathbf{conv}\, X$, is the set of all convex combinations of points in X:

$$\mathbf{conv}\, X = \{\alpha_1 \boldsymbol{x}_1 + \cdots + \alpha_k \boldsymbol{x}_k \,|\, i = 1 \ldots k,\ \alpha_i \geq 0,\ \textstyle\sum_i \alpha_i = 1\}$$

The convex hull is the smallest convex set that contains X.

The convex hulls of two sets



Convexity of Functions

A function $f : X \to \mathbb{R}$ is convex if X is convex and:

1. The ***epigraph*** of f is a ***convex set***

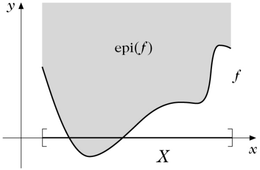
$$\mathbf{epi}\, f = \{(\boldsymbol{x}, t) \,|\, \boldsymbol{x} \in X, f(\boldsymbol{x}) \leq t\}$$

2. ***Jensen's Inequality:*** For all $\boldsymbol{x}, \boldsymbol{y} \in X$ and $0 \leq \alpha \leq 1$

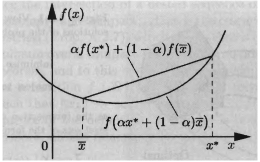
$$f(\alpha \boldsymbol{x} + (1-\alpha)\boldsymbol{y}) \leq \alpha f(\boldsymbol{x}) + (1-\alpha) f(\boldsymbol{y})$$

3. The function f is twice differentiable and its ***Hessian*** $\nabla^2 f(\boldsymbol{x})$ is ***positive semi-definite*** for all $\boldsymbol{x} \in X$.
(This is explained in more detail in the next lecture.)

Epigraph of a (non-convex) function



Jensen's Inequality



The conditions 1-3 are equivalent. If one is true, the others must be true as well.

Subgradient and Subdifferential

Let X a ***nonempty, open, convex*** subset of $\mathbb{R}^n$ and let $f : X \to \mathbb{R}$ ***differentiable everywhere.***

A ***subgradient*** of f at $\boldsymbol{x} \in X$ is a vector $\boldsymbol{g} \in \mathbb{R}^n$ for which

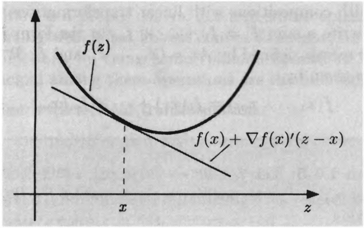
$$f(\boldsymbol{z}) \geq f(\boldsymbol{x}) + \boldsymbol{g}^\top(\boldsymbol{z} - \boldsymbol{x}), \qquad \forall \boldsymbol{x}, \boldsymbol{z} \in X\,. \tag{1}$$

If f is convex and differentiable, then the ***gradient*** $\nabla f(\boldsymbol{x})$ ***is the only subgradient.*** For ***strictly convex*** f, the above inequality is strict whenever $\boldsymbol{x} \neq \boldsymbol{z}$.

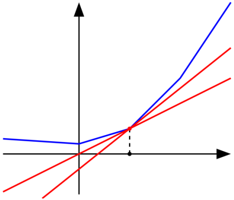
What if f is not everywhere differentiable?

- A convex function must be continuous, but is not necessarily differentiable at every point.
- For convex functions, ***multiple subgradients*** exist for $\boldsymbol{x}$ ***where*** f ***is non-differentiable.***
- The ***subdifferential*** $\partial f(\boldsymbol{x})$ is the ***set of all subgradients*** $\boldsymbol{g} \in \partial f(\boldsymbol{x})$ that hold Equation 1.

Subgradient



Subgradients of non-diff'able function



Known convex functions

- $\boldsymbol{a}^\top \boldsymbol{x} + b$
- $x \log x$ for $x > 0$
- $\boldsymbol{x}^\top \mathbf{P} \boldsymbol{x}$ for $\mathbf{P} \succeq 0$
- $\|\boldsymbol{x}\|$ for any norm
- $\max(x_1, \ldots, x_n) = \|\boldsymbol{x}\|_\infty$

Known concave functions

- x^p for $0 \leq p \leq 1$
- $\log x$ for $x > 0$
- $\sqrt{xy}$
- $\boldsymbol{x}^\top \mathbf{P} \boldsymbol{x}$ for $\mathbf{P} \preceq 0$
- $\min(x_1, \ldots, x_n)$

Composition of convex/concave functions

$f(g_1(\boldsymbol{x}), g_2(\boldsymbol{x}), \ldots)$ is convex if f ***is convex*** and for ***all arguments*** i ***either***

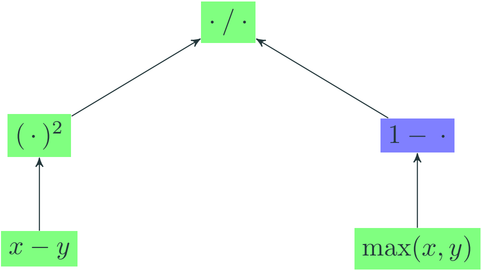
- f increases in argument i and g_i is convex
- f decreases in argument i and g_i is concave
- g_i is linear

Known convex / concave primitive functions.

Composition rules that preserve convexity.

Attention! The composition rules are ***sufficient*** but ***not necessary*** for convexity.

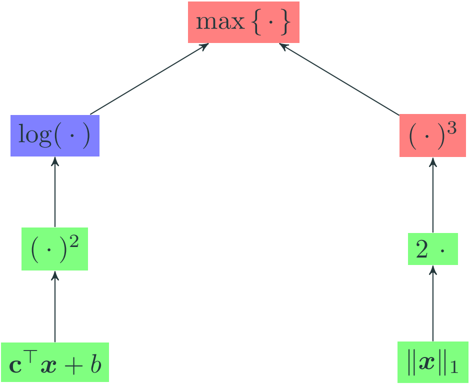
$$\min_{x,y} \frac{(x-y)^2}{1-\max(x,y)}$$



Green: convex or affine, Blue: concave, Red: neither

Convexity Quiz II

$$\min_{\boldsymbol{x}} \max\left\{\log\left((\boldsymbol{c}^\top \boldsymbol{x} + b)^2\right), (2\|\boldsymbol{x}\|_1)^3\right\}$$



Green: convex or affine, Blue: concave, Red: neither

Gradient Descent

Direction of Steepest Descent

- The 1st degree Taylor Expansion ***linearizes the function*** $f : \mathbb{R}^n \to \mathbb{R}$ at the selected point $\boldsymbol{x}$:

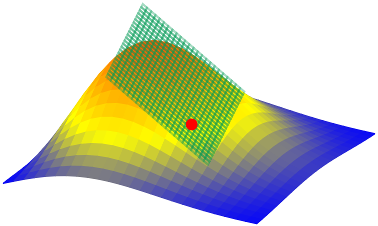
$$\tau_{\boldsymbol{x}}^{1}(\boldsymbol{y}) = f(\boldsymbol{x}) + (\boldsymbol{y} - \boldsymbol{x})^{\top} \nabla f(\boldsymbol{x})$$

- In which direction can we ***step downwards as far as possible*** on the tangent plane of the 1st Degree Taylor Expansion (for a step of length 1)?

$$-\frac{\nabla f(\boldsymbol{x})}{\|\nabla f(\boldsymbol{x})\|_2} = \underset{\boldsymbol{h} \in \mathbb{R}^n, \|\boldsymbol{h}\|_2 = 1}{\arg\min}\ \boldsymbol{h}^{\top} \nabla f(\boldsymbol{x})$$

- The ***steepest descent*** is in the direction of the ***negative gradient*** $-\nabla f(\boldsymbol{x})$

Tangent Plane of the 1st Degree Taylor Expansion



Proof: The **Cauchy-Schwarz Inequality** tells us that $|\boldsymbol{h}^{\top} \nabla f(\boldsymbol{x})| = \|\boldsymbol{h}^{\top} \nabla f(\boldsymbol{x})\|_2 \leq \|\boldsymbol{h}\|_2 \|\nabla f(\boldsymbol{x})\|_2$. The right-hand-side is fixed since $\|\boldsymbol{h}\|_2 = 1$. The left-hand-side is maximized when equality is achieved. Equality is achieved for $\boldsymbol{h}^* = \frac{\nabla f(\boldsymbol{x})}{\|\nabla f(\boldsymbol{x})\|_2}$. Take $-\boldsymbol{h}^*$ to minimize.

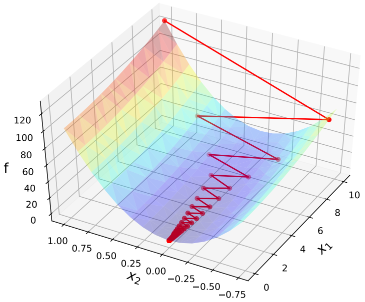
- ***Iterative method*** starting at an initial point $\boldsymbol{x}^{(0)}$
- Step to the next point $\boldsymbol{x}^{(k+1)}$ in the ***direction of the negative gradient***

$$\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} - \nabla f(\boldsymbol{x}^{(k)})$$

- Repeat until $\|\nabla f(\boldsymbol{x}^{(k)})\| < \epsilon$ for a chosen ϵ
- **But:** No convergence is guaranteed.
  For convergence, an additional ***line search*** is required.

Line Search

- Take the descent step direction $\boldsymbol{d} = -\nabla f(\boldsymbol{x})$
- Select the step length α as $\min_{\alpha \geq 0} f(\boldsymbol{x} + \alpha \boldsymbol{d})$
- In practice, α is selected with heuristics



Gradient Descent for
$f(\boldsymbol{x}) = \frac{1}{2}(x_1)^2 + 5(x_2)^2$

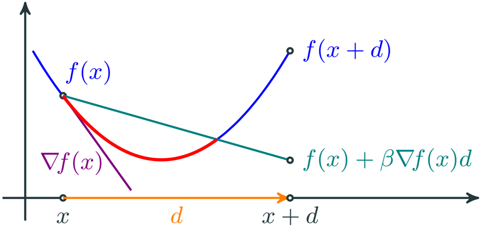
Backtracking Line Search [Armijo66]

- Heuristic line search (not exact), but simple and efficient
- Start with a ***step direction*** $\boldsymbol{d}$
- ***Iteratively reduce*** the ***step length factor*** α. A common rule is $\alpha \leftarrow p\alpha$ with $p = 0.5$.
- Stop when a minumum “descent steepness” is reached (Armijo Condition)

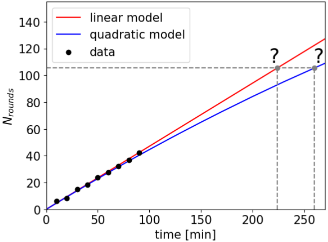
$$f(\boldsymbol{x} + \alpha \boldsymbol{d}) \leq f(\boldsymbol{x}) + \beta [\nabla f(\boldsymbol{x})]^\top (\alpha \boldsymbol{d})$$

Choose $\beta \in (0, 1)$. A common choice is $\beta = 10^{-4}$.
$\beta \approx 0$: Any descent is valid (must be only minimally downwards) $\rightarrow$ Big nearly horizontal steps allowed
$\beta \approx 1$: The descent needs to be nearly as steep as $-\nabla f(\boldsymbol{x})$ $\rightarrow$ Step sizes can become very small

procedure LINESEARCH($f, \boldsymbol{x}, \boldsymbol{d}, p, \beta$)
    $\alpha \leftarrow 1$
    while $f(\boldsymbol{x} + \alpha \boldsymbol{d}) > f(\boldsymbol{x}) + \beta [\nabla f(\boldsymbol{x})]^\top (\alpha \boldsymbol{d})$ **do**
        $\alpha \leftarrow p\alpha$
    end while
    return α
end procedure



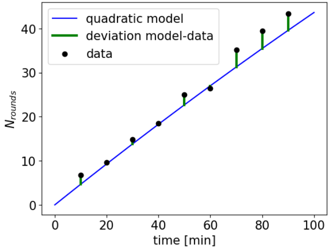
Model Fitting: Empirical Risk Minimization

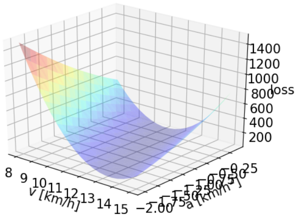


- Suppose you are training for a marathon in a stadium
- You record your running speed to track your training progress
- You do not run the full 42 km (105 rounds) during training. Can you ***predict how long the full marathon would have taken***?

1. **Select a model class for the distance prediction**
   - Constant speed: $p(t) = vt$
   - With slowdown: $p(t) = vt - at^2$
   - These are ***parametric models***. They define a fixed number of model parameters (initial speed v and deceleration a).
2. **Fit the model parameters**
   - Finding "good" model parameters is called ***model fitting***.
   - ***Empirical risk minimzation*** is commonly used for model fitting.

Minimize the difference between predictions and observations





- Assume the model class with slowdown: $p(t) = vt - at^2$
- The dataset with times and corresponding distances is

$$D = \{(t^1, d^1), (t^2, d^2), \dots\}$$

- For this example, we decide to use a ***quadratic prediction error*** for each observation (t^i, d^i)
- The ***loss function*** returns the overall prediction error for a combination of model parameters v, a

$$l(v, a) = \sum_i \left(d^i - p(t^i)\right)^2$$

- ***Minimize the loss function*** to find the "best" model parameters

$$(v^*, a^*) = \arg\min_{v,a} \; l(v, a)$$

Summary of what you learned today

- ***Optimization***, some historical applications and its importance in many domains of science and engineering
- Recap of some basic notions from ***Analysis***
- ***Convexity and Convex Functions***
  - What it is
  - Why it is important
  - How to detect it
- ***Gradient Descent***
  - A universal solution method for differentiable strictly convex functions
  - ***Line Search*** to guarantee convergence of Gradient Descent
- ***Empirical Risk Minimization***, a central optimization problem in Machine Learning

That’s it for today.

See you next week for Lecture 2 on the

Newton Method

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Image Sources

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